INTEGRATION BEE 2025

Fase Escrita – Prova I

1. $$\int_{2.025}^{5.025} \frac{d}{dx}\left(x + \cos(2\pi x)^3\right) dx = 3$$

2. $$-\int \sin^5(x)dx = \cos(x) - \frac{2\cos^3(x)}{3} + \frac{\cos^5(x)}{5}$$

3. $$\int_1^{\infty} \frac{1}{x + x^{2025}}dx = \frac{\log(2)}{2024}$$

4. $$\int_1^{\infty} \frac{\log(1+x)}{x\sqrt{x}}dx = 2\log(2) + \pi$$

5. $$\int_1^{\infty} \frac{1}{(x+8)\sqrt{x-1}}dx = \frac{\pi}{3}$$

6. $$\int_0^{2025} (x - \lfloor x \rfloor)dx = \frac{2025}{2}$$

7. $$\int_0^{\pi/2} \frac{(\sin x)^{2025}}{(\sin x)^{2025} + (\cos x)^{2025}}dx = \frac{\pi}{4}$$

8. $$\int \frac{1}{x(4 - \log^2(x))}dx = \frac{1}{4}\log\left|\frac{2 + \log(x)}{2 - \log(x)}\right|$$

9. $$\int_{-\infty}^{+\infty} \frac{x + 2x^3}{(x^2 + x^6 + 1)^{3/2}}dx = 0$$

10. $$\int 2\cosh(x)\cos^2(x)dx = \sinh(x) + \frac{\sinh(x)\cos(2x) + 2\cosh(x)\sin(2x)}{5}$$

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Fase Escrita – Prova II

1. $$\int e^x e^{e^x} e^{e^{e^x}} e^{e^{e^{e^x}}} e^{e^{e^{e^{e^x}}}} \, dx = e^{e^{e^{e^{e^x}}}}$$

2. $$\int_1^2 \log^2(x) dx = 2\left(\log^2(2) - 2\log(2) + 1\right)$$

3. $$\int_0^\infty \frac{1}{1+e^{2025x}} dx = \frac{\log(2)}{2025}$$

4. $$\int_{-1}^1 \frac{x^5 + 3x^{11} + x^{12} + 2x^{18} + x^{24}}{(x^6 + x^{12})^2} dx = 2$$

5. $$\int_0^2 \sqrt{1 - \frac{x^2}{4}} dx = \frac{\pi}{2}$$

6. $$\int \frac{1}{\log(x)} + \log\left(\log(x)\right) dx = x\log(\log(x))$$

7. $$\int_0^{+\infty} \frac{\log(x)}{1+x^2} dx = 0$$

8. $$\int \frac{x^{2025} - 1}{x^{405} - 1} dx = \frac{x^{1621}}{1621} + \frac{x^{1216}}{1216} + \frac{x^{811}}{811} + \frac{x^{406}}{406} + x$$

9. $$\int \frac{\cos^3(x)}{\sqrt{1+\sin^2(x)}} dx = \frac{3}{2}\operatorname{arcsinh}(\sin(x)) - \frac{1}{2}\sin(x)\sqrt{1+\sin^2(x)}$$

10. $$\int \frac{2x^3 - 1}{x(x^3+1)(1+(\log(x^3+1) - \log(x))^2)} dx = \arctan\left(\log\left(x^2 + \frac{1}{x}\right)\right)$$